Geographic Variation in the Black-White Life Expectancy Gap in Wisconsin, 2005-2017

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Abstract

Racial disparities in life expectancy persist across the United States, but state-level averages can mask important geographic heterogeneity. This paper examines the Black-White life expectancy gap in Wisconsin from 2005 to 2017, disaggregating the state into three regions: Milwaukee County, Dane County (Madison), and the rest of the state. Using individual death records and CDC bridged-race population estimates, I construct abridged life tables for non-Hispanic Black and White populations using three-year pooled data. I partially replicate the statewide findings of Roberts et al. (2019) and extend the analysis to reveal substantial geographic variation. In the most recent period (2015--2017), the male Black-White gap ranges from 3.8 years in the rest of Wisconsin to 9.2 years in Milwaukee County, compared to a statewide average of 7.9 years. Dane County shows the most alarming trend: the male gap widened by 3.2 years over the study period, driven by a 2.4-year decline in Black male life expectancy. These findings underscore the importance of sub-state geographic analysis for understanding and addressing racial health disparities.

Keywords: life expectancy, racial disparities, Wisconsin, geographic variation, mortality

1. Introduction

The Black-White gap in life expectancy is one of the most persistent and well-documented health disparities in the United States. Although this gap has narrowed nationally over the past several decades, substantial variation remains across states and localities (Harper et al. 2014; Riddell et al. 2018). State-level analyses, while informative, can obscure important within-state heterogeneity driven by residential segregation, differential access to health care, economic inequality, and other social determinants of health.

Wisconsin consistently ranks among the worst states for Black-White disparities across a range of social and economic indicators, including life expectancy. The Milwaukee metropolitan area, in particular, has been repeatedly identified as one of the most racially segregated in the nation. Roberts et al. (2019) documented a persistent statewide Black-White life expectancy gap in Wisconsin that showed little improvement between 1999 and 2016, in contrast to national trends of convergence. Their analysis, however, was conducted at the state level and did not examine geographic variation within Wisconsin.

This paper extends the analysis of Roberts et al. (2019) in two ways. First, I replicate their statewide life expectancy trends to validate the analytic approach. Second, I disaggregate the state into three meaningful regions: Milwaukee County, which contains the majority of Wisconsin's Black population and is among the most racially segregated metropolitan areas in the country; Dane County (Madison), a growing metropolitan area with a smaller but increasing Black population; and the rest of the state. This geographic decomposition reveals striking differences in both the magnitude of the Black-White gap and its trajectory over time.

2. Data and Methods

2.1 Data Sources

Mortality data come from individual death records for all deaths in Wisconsin from 2005 to 2017. Each record includes year of death, county of residence (FIPS code), age, sex, race, and Hispanic origin. I restrict the analysis to deaths among non-Hispanic Black and non-Hispanic White individuals, identified using the NCHS Hispanic origin coding (codes < 200 indicating non-Hispanic origin). The data contain 614,954 deaths meeting these criteria.

Population denominators come from the CDC's bridged-race population estimates for Wisconsin counties, 2005--2017, by single year of age (0--89, 90+), sex, and race. A limitation of the population data is that Hispanic origin is not separately identified (origin = 9 for all records). The population denominators therefore include Hispanic individuals within each racial category, which may slightly inflate the denominators for non-Hispanic populations. This is a minor concern for Black populations in Wisconsin, where the Hispanic share is small.

Geographic regions are defined using county FIPS codes: Milwaukee County (FIPS 55079), Dane County (FIPS 55025), and the rest of Wisconsin (all other 55xxx codes). Milwaukee County is home to approximately 60% of Wisconsin's Black population, while Dane County accounts for roughly 8%.

2.2 Life Table Construction

I construct abridged period life tables following Chiang (1968) and Preston et al. (2001). Deaths and populations are aggregated into 19 age groups: <1, 1--4, 5--9, ..., 80--84, 85+. To stabilize rates for smaller populations, I pool data over three-year windows centered on each year from 2006 to 2016 (e.g., the "2010" estimate uses data from 2009--2011). Both deaths and population are summed across the three-year window, so the resulting rates represent average annual age-specific mortality.

The fraction of the interval lived by those dying in the interval (nax) is calculated using Coale-Demeny formulas for infants (age < 1) based on the level of infant mortality and sex, as presented in Preston et al. (2001, Table 3.3). For children aged 1--4, nax is similarly derived from the infant mortality rate. For ages 5--84, nax is set to half the interval width (2.5 years for 5-year groups). For the open-ended interval (85+), nax is set to the reciprocal of the age-specific mortality rate (1/M), which estimates mean remaining lifetime conditional on surviving to age 85.

The Black-White gap is computed as the difference in life expectancy at birth (e0) between the non-Hispanic White and non-Hispanic Black populations for each sex-geography-period combination.

2.3 Replication Strategy

I first replicate the statewide life expectancy estimates from Roberts et al. (2019) for validation. My estimates are expected to differ slightly because I restrict to 2005--2017 rather than 1999--2016, and minor differences in age-coding or Hispanic-origin filtering may exist. A close match (within ~0.5 years) provides confidence in the analytic pipeline before extending to sub-state geographies.

3. Results

3.1 Replication of Statewide Trends

Table 1 compares my statewide estimates for the 2014--2016 period (center year 2015) with those reported in Roberts et al. (2019). My estimates are within 0.1 to 0.4 years of the published values across all race-sex groups. The small discrepancies likely reflect differences in Hispanic-origin filtering and the exact population denominators used. The Black-White gap in my estimates is 7.2 years for males and 5.6 years for females, compared to 7.3 and 5.6 years in Roberts et al.

Table 1. Replication check: statewide life expectancy (years) for 2014--2016, compared with Roberts et al. (2019).

Group	Roberts et al.	This Paper	Difference
White Males	77.75	78.06	+0.31
Black Males	70.41	70.84	+0.43
White Females	82.15	82.24	+0.09
Black Females	76.54	76.67	+0.13
Male B-W Gap	7.34	7.22	-0.12
Female B-W Gap	5.61	5.57	-0.04

Figures 1 and 2 show statewide life expectancy trends for males and females, respectively. Among males, White life expectancy was essentially flat over the period (77.4 to 78.0 years), while Black male life expectancy rose from 69.8 to a peak of 71.6 years around 2012 before declining to 70.0 years by 2016. This reversal widened the male gap from approximately 6.5 years at its narrowest (2012) to 7.9 years at the end of the study period. Among females, trends were more stable, with both groups showing modest gains and the gap fluctuating between 5.6 and 6.2 years.

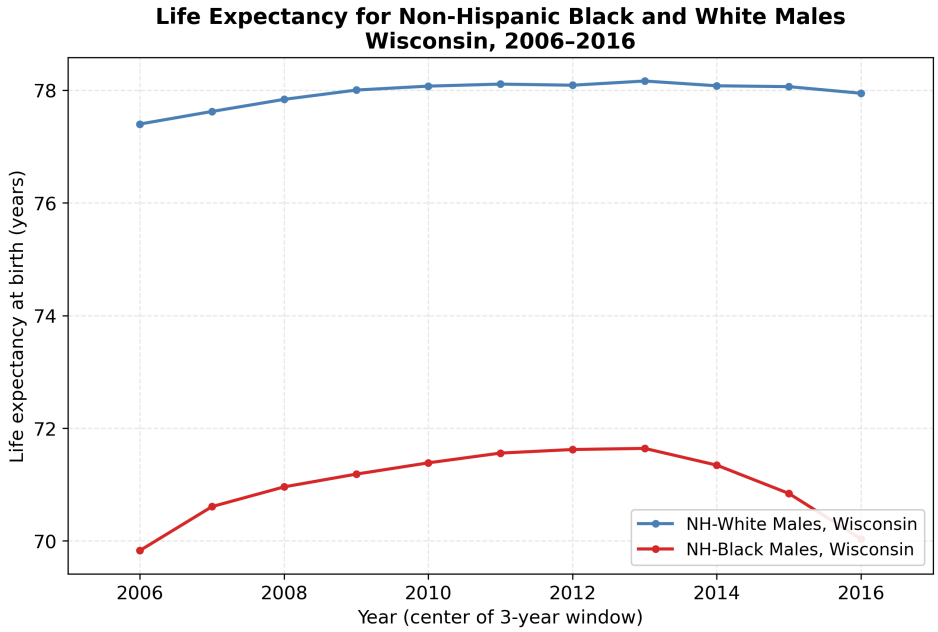

Figure 1. Life expectancy at birth for non-Hispanic Black and White males, Wisconsin statewide, 2006--2016 (3-year pooled estimates).

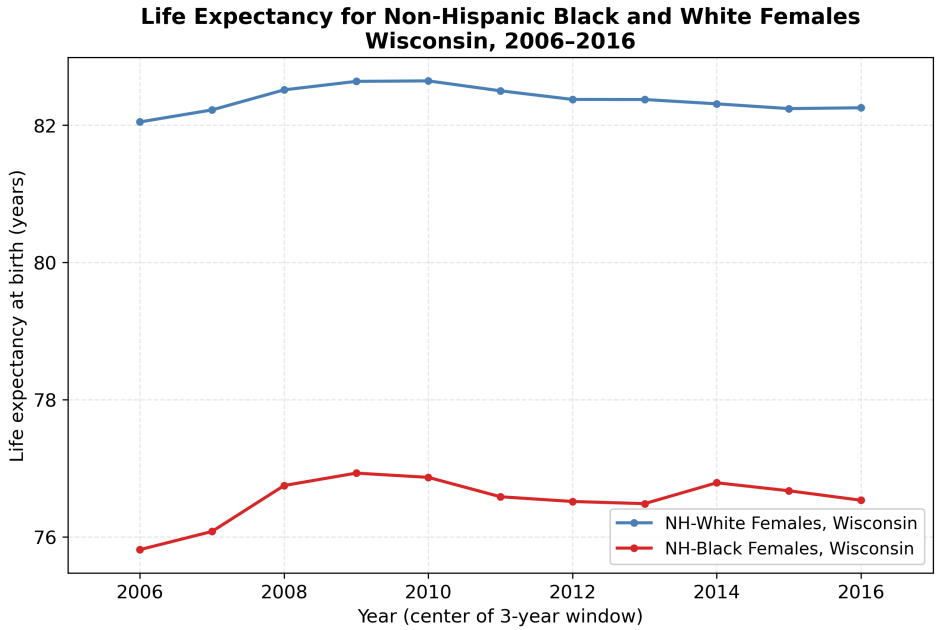

Figure 2. Life expectancy at birth for non-Hispanic Black and White females, Wisconsin statewide, 2006--2016 (3-year pooled estimates).

3.2 Geographic Variation in the Black-White Gap

Table 2 presents life expectancy and the Black-White gap for the most recent period (2015--2017) across the four geographic units. The results reveal substantial heterogeneity masked by the statewide average.

Table 2. Life expectancy at birth (years) by race, sex, and geography, 2015--2017 (3-year pooled).

Geography	Sex	Black LE	White LE	B-W Gap
Dane	Female	78.30	83.73	5.43
Dane	Male	71.71	80.21	8.50
Milwaukee	Female	75.95	82.05	6.10
Milwaukee	Male	68.25	77.48	9.23
Rest of WI	Female	77.81	82.13	4.32
Rest of WI	Male	73.94	77.78	3.84
Statewide	Female	76.53	82.25	5.72
Statewide	Male	70.03	77.95	7.92

For males, the Black-White gap ranges from 3.8 years in the rest of Wisconsin to 9.2 years in Milwaukee County -- a nearly 2.5-fold difference. Dane County's male gap (8.5 years) is nearly as large as Milwaukee's, despite Dane County's substantially higher overall life expectancy. This reflects a dual pattern: White males in Dane County enjoy the highest life expectancy of any group (80.2 years), but Black males in Dane County do not share proportionally in this advantage (71.7 years). Milwaukee has the lowest Black male life expectancy of any region (68.3 years), which is 5.7 years below the rest-of-state average for Black males.

For females, the geographic pattern is less dramatic but still notable. The gap ranges from 4.3 years (rest of Wisconsin) to 6.1 years (Milwaukee). Dane County shows the smallest female gap (5.4 years), driven by relatively high Black female life expectancy there (78.3 years, the highest of any region). A caveat is warranted: Dane County's Black population is small relative to the other regions (roughly 50,000 over the 3-year window, with approximately 260 male and 158 female Black deaths), which limits the precision of these estimates.

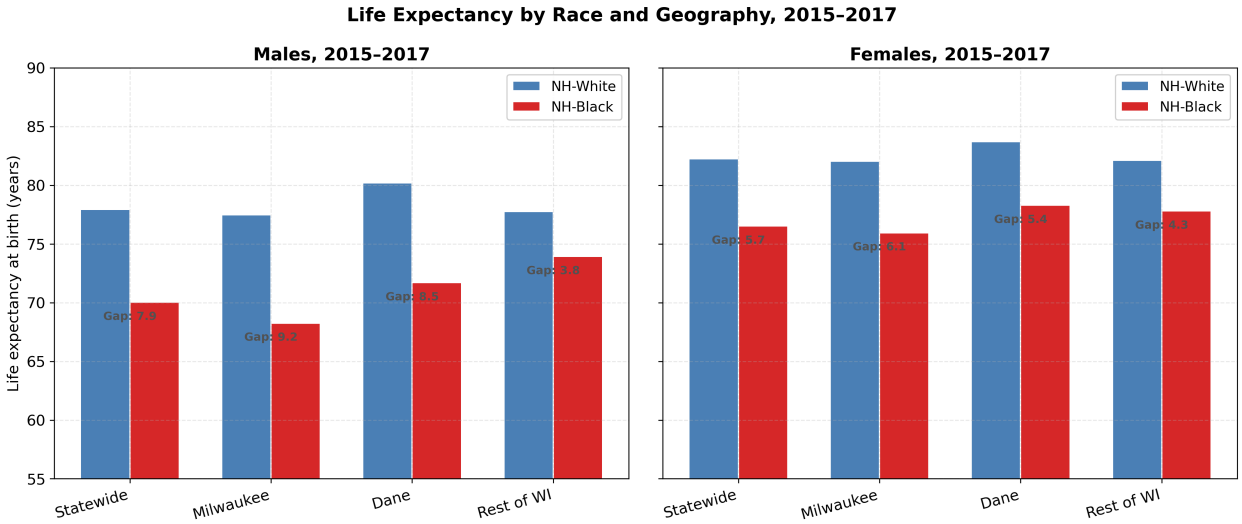

Figure 3. Life expectancy at birth by race and geography, males (left) and females (right), 2015--2017.

3.3 Trends in the Gap by Geography

Figures 4 and 5 plot the Black-White gap over time for each geography. The most striking finding is the divergent trajectory of the male gap across regions. In Milwaukee, the male gap narrowed from 8.6 years (2006) to a low of 7.1 years (2010) before widening sharply to 9.2 years by 2016. In Dane County, the male gap nearly doubled from 5.3 years (2006) to 8.5 years (2016). In the rest of Wisconsin, the gap remained comparatively small but also widened, from 2.1 to 3.8 years.

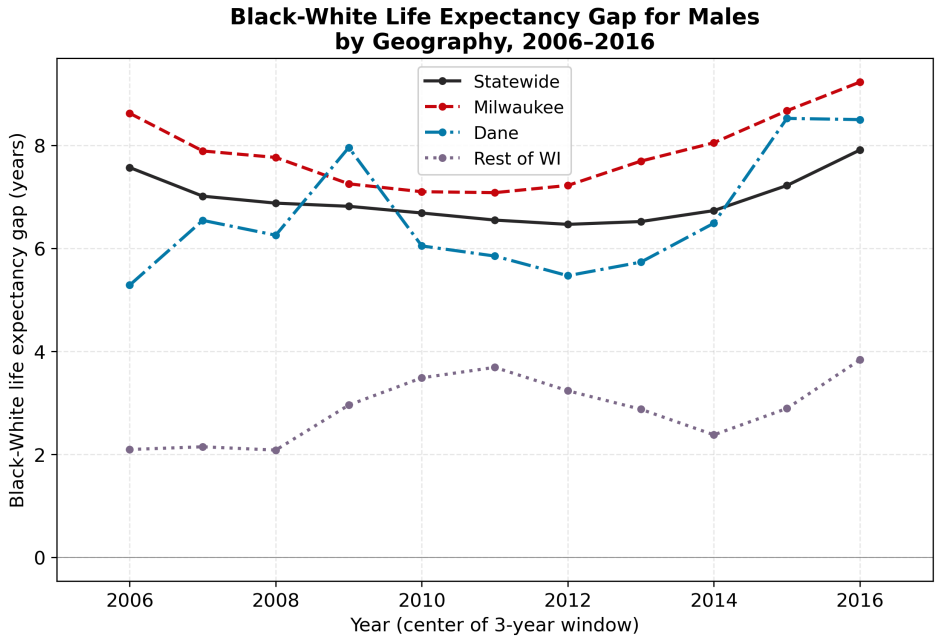

Figure 4. Black-White life expectancy gap for males by geography, 2006--2016 (3-year pooled estimates).

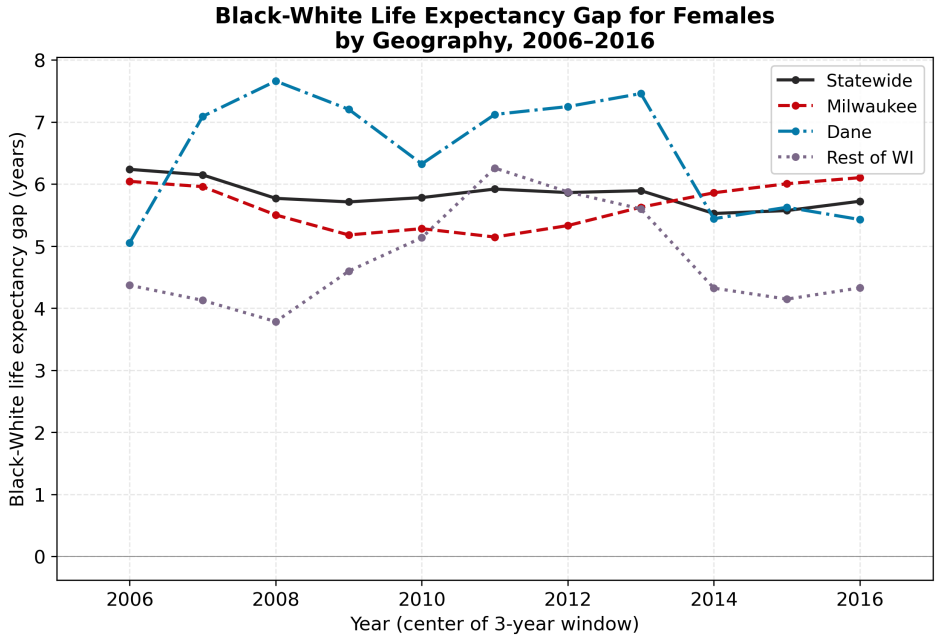

Figure 5. Black-White life expectancy gap for females by geography, 2006--2016 (3-year pooled estimates).

For females, trends were more stable across geographies. The statewide female gap narrowed slightly from 6.2 to 5.7 years. Milwaukee's female gap remained roughly constant near 6 years. Dane County showed considerable volatility in the female gap -- likely reflecting the small Black female population -- but ended the period at 5.4 years, similar to where it began.

3.4 Decomposing the Change

Table 3 decomposes the change in the Black-White gap between the earliest (2005--2007) and latest (2015--2017) periods into changes in White and Black life expectancy. This decomposition is informative because a widening gap could reflect White gains, Black losses, or both.

Table 3. Change in the Black-White life expectancy gap (years) between 2005--2007 and 2015--2017, with decomposition into White and Black LE changes.

Geography	Sex	Gap (05--07)	Gap (15--17)	Change	Delta White LE	Delta Black LE
Statewide	Male	7.57	7.91	0.34	0.55	0.20
Statewide	Female	6.24	5.72	-0.52	0.20	0.72
Milwaukee	Male	8.62	9.23	0.61	1.13	0.52
Milwaukee	Female	6.04	6.10	0.06	0.77	0.71
Dane	Male	5.29	8.50	3.21	0.85	-2.37
Dane	Female	5.05	5.43	0.38	0.46	0.08
Rest of WI	Male	2.10	3.84	1.74	0.41	-1.33
Rest of WI	Female	4.37	4.33	-0.04	0.07	0.12

The most alarming finding is in Dane County, where the male gap widened by 3.2 years -- the largest increase of any region-sex combination. This was driven primarily by a 2.4-year decline in Black male life expectancy (from 74.1 to 71.7 years), while White male life expectancy increased by 0.9 years. In Milwaukee, the male gap widened by 0.6 years, with both groups gaining but White males gaining faster (+1.1 years vs. +0.5 years for Black males). In the rest of Wisconsin, Black male life expectancy declined by 1.3 years, widening the gap by 1.7 years.

Among females, the statewide gap narrowed by 0.5 years, driven by faster improvement among Black females (+0.7 years) compared to White females (+0.2 years). This pattern was not uniform across geographies: Milwaukee showed essentially no change, while the rest of Wisconsin also showed negligible change in the female gap.

4. Discussion

This analysis reveals three key findings about the Black-White life expectancy gap in Wisconsin. First, the statewide gap masks substantial geographic variation: the male gap in Milwaukee (9.2 years) is more than double that in the rest of the state (3.8 years). Second, the gap widened in all regions for males between 2005--2007 and 2015--2017, with the most dramatic deterioration in Dane County. Third, the widening gap reflects not just differential rates of improvement but, in some regions, actual declines in Black male life expectancy.

The large and growing gap in Dane County is particularly noteworthy. Dane County (home to Madison) is often perceived as a progressive, high-opportunity community with strong public institutions and a robust economy. Yet the Black-White male life expectancy gap there (8.5 years) is nearly as large as Milwaukee's (9.2 years), and it widened more rapidly over the study period. This suggests that favorable average conditions do not automatically translate into equitable health outcomes, and that the social determinants driving mortality disparities may operate through mechanisms -- such as residential

segregation within the county, differential incarceration rates, or concentrated poverty among Black residents -- that are not captured by county-level aggregates.

The decline in Black male life expectancy in several regions during the latter part of the study period is consistent with emerging national trends linked to the opioid epidemic, increases in homicide, and rising "deaths of despair." Further research using cause-of-death decomposition (e.g., Arriaga 1984) could identify which specific causes are driving these reversals.

Several limitations should be noted. The population denominators do not separately identify Hispanic origin, which may introduce a small upward bias in population counts for non-Hispanic groups. Dane County's Black population, while growing, remains relatively small; the 3-year pooling mitigates but does not eliminate instability in the estimates. The analysis is ecological in nature -- geographic comparisons cannot identify the individual-level mechanisms driving disparities. The analysis covers 2005--2017 and does not capture more recent trends, including the COVID-19 pandemic, which disproportionately affected Black life expectancy nationwide. Finally, this analysis does not decompose the gap by cause of death, which would help identify targeted intervention points; such decomposition is a priority for future work.

5. Conclusion

Statewide life expectancy statistics, while valuable, can obscure the geographic concentration of racial health disparities. In Wisconsin, the Black-White life expectancy gap varies by a factor of nearly 2.5 across regions, and the trajectories of the gap are diverging. Milwaukee's large and persistent gap reflects deep structural inequalities in one of America's most segregated cities. Dane County's rapidly widening gap is a warning that health equity requires intentional investment even in communities with otherwise favorable conditions. Policy responses should be tailored to local contexts rather than relying on statewide averages that mask the communities most in need of intervention.

Appendix: Geographic Reference

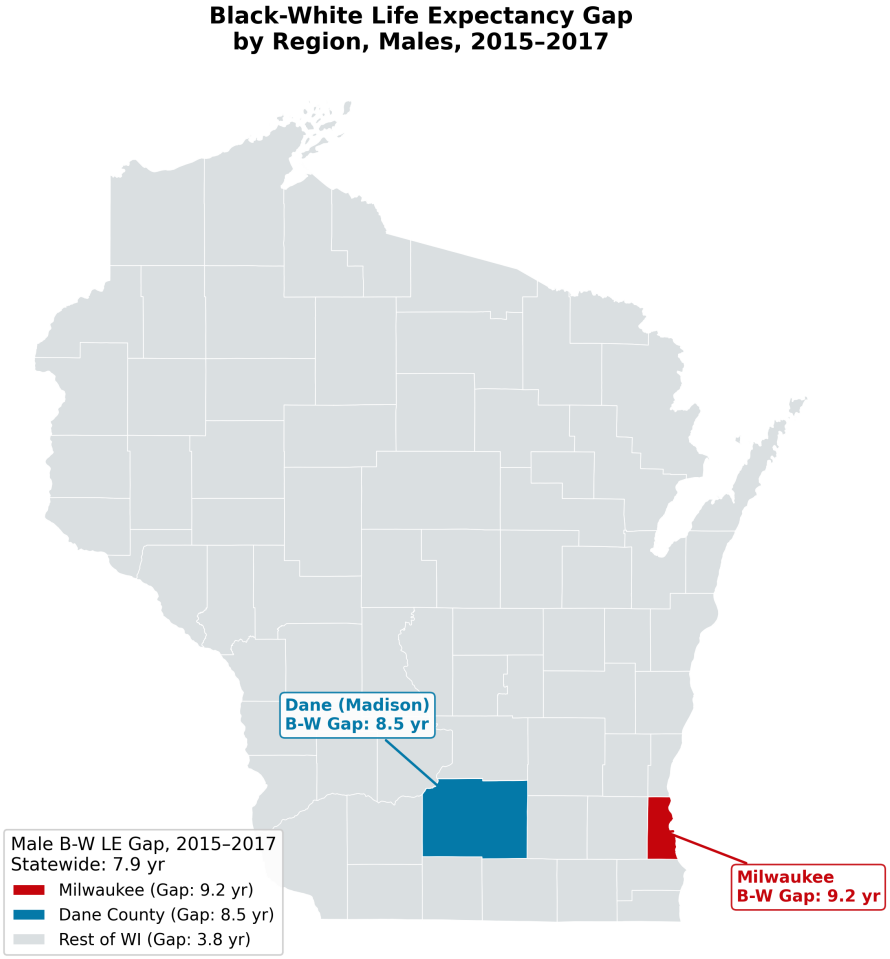

Figure A1. Wisconsin counties with Milwaukee, Dane, and rest-of-state regions highlighted. Annotations show male Black-White life expectancy gap for 2015--2017.

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